SSPNet: Spatial–Spectral Perception Network for Mineral Hyperspectral Image Classification

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***Abstract*—Unlike general scenes, mineral hyperspectral images (HSIs) often exhibit similar spatial and spectral characteristics across different mines, making traditional classification methods less effective due to compromised robustness. To address this, we propose a spatial–spectral perception network (SSPNet) for mineral HSI classification. This approach divides spatial–spectral feature extraction into two stages. In the spatial feature perception stage, we introduce a spatial–frequency perceptron (SFP) that maps 3-D spatial features into low- and high-frequency domains. We then apply triple-cross-attention (TCA) to each frequency domain to better differentiate spatial features of similar mines. In the spectral perception stage, we design a spectral linear perceptron (SLP) using absolute linear attention (ALA), which captures fine-grained spectral differences by establishing internal relationships between spectral features through absolute positional weighting. This enables effective separation of similar spectra for final classification. Extensive experiments on three publicly available mineral HSI datasets and one agricultural hyperspectral dataset show that our method outperforms popular alternatives in both effectiveness and robustness. The open-source code can be accessed at https://github.com/WUTCM-Lab/SSPNet**

***Index Terms*—Attention mechanism, mineral hyperspectral image (HSI) classification, position encoding, spatial–spectral feature extraction.**

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I. Introduction

HYPERSPECTRAL imaging is a powerful remote sensing technology that captures images across hundreds of contiguous spectral bands, providing detailed spectral information beyond the capabilities of conventional RGB cameras. This enables the detection of subtle variations in the chemical composition, structure, and condition of materials, making hyperspectral imaging an invaluable tool in many fields such as environmental monitoring, mineral exploration, and agriculture. In recent years, hyperspectral image (HSI) classification has emerged as a critical task in remote sensing. The goal of HSI classification is to categorize each pixel in the image into predefined classes based on its spectral signature. However, due to the high dimensionality of hyperspectral data (often containing hundreds of spectral bands), this task presents unique challenges, both from a computational and analytical perspective. Currently, researchers typically apply dimensionality reduction techniques to high-dimensional HSIs. Common dimensionality reduction methods include principal component analysis (PCA) [1], independent component analysis (ICA) [2], linear discriminant analysis (LDA) [3], and locally linear embedding (LLE) [4]. These strategies help reduce feature redundancy along the spectral dimension and significantly decrease the computational burden during model training. Additionally, many researchers are focused on developing effective hyperspectral feature compression methods. Christophe et al. [5] optimized a wavelet-based compression system for HSIs by defining an optimal 3-D wavelet decomposition and adapting zero tree coding methods (EZW and SPIHT), demonstrating improved performance and lower complexity compared to classical methods. Du and Chang [6] presented an unsupervised fully constrained least-squares (UFCLSs) linear spectral mixture analysis-based compression technique for hyperspectral target detection and classification, which preserves vital information in fractional abundance images, achieving high compression ratios with minimal impact on image analysis performance.

Early HSI classification methods primarily relied on traditional machine learning algorithms. These methods typically use the spectral information of each pixel in an HSI as input features, which are then classified using a classifier. Common traditional machine learning methods include support vector machines (SVMs) [7], decision trees [8], k-nearest neighbors (k-NNs) [9], and random forests (RFs) [10], among others. For instance, SVMs have been widely applied to HSI classification due to their excellent performance in high-dimensional spaces. Guo et al. [11] proposed spectrally weighted kernels for

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SVM HSI classification to improve performance by optimizing band-specific weights, with experiments showing that gradient descent-based weight learning slightly outperforms relevance-based methods. By constructing a hyperplane to separate pixels of different classes, SVMs effectively address classification problems in high-dimensional feature spaces. In addition, Ham et al. [12] proposed a new binary hierarchical multi-classifier system using bagging and adaptive random subspace feature selection to improve the generalization of hyperspectral data classification, especially with limited training data, outperforming traditional methods in experiments using NASA hyperspectral datasets. However, traditional machine learning methods often rely on manually designed features, and their performance is limited when dealing with complex data.

With the rapid development of deep learning technologies, deep neural networks have gradually achieved significant success in HSI classification [13]. Deep learning methods, especially convolutional neural networks (CNNs), are capable of automatically learning effective feature representations from raw data, eliminating the need for manual feature extraction [14]. This enables deep learning methods to fully exploit the rich spectral and spatial information within HSIs, overcoming some of the limitations of traditional machine learning approaches [15]. CNNs construct deep network models through multiple convolutional layers, pooling layers, and fully connected layers, allowing them to automatically extract meaningful feature representations from each pixel and its surrounding neighborhood in an HSI [16]. In addition to CNNs, other deep learning techniques, such as recurrent neural networks (RNNs) [17], generative adversarial networks (GANs) [18], and multihead self-attention mechanisms [19], have also been gradually applied to HSI classification, further enhancing classification accuracy and robustness [20].

In recent studies, attention mechanisms have shown remarkable performance in extracting spatial–spectral features. To address the limitations of CNNs in establishing dependencies among long sequence data, researchers have introduced a cross-attention mechanism. Yang et al. [21] proposed an adaptive cross-attention driven spatial–spectral graph convolutional network (ACSS-GCN). The network extracts spatial and spectral features by Sa-GCN and Se-GCN and fuses these features using GCAFM to suppress noise interference. Cai and Wei [22] proposed a novel cross-attention mechanism and graph convolution integration algorithm, which used a cross (horizontal and vertical directions) attention algorithm to allocate weights jointly based on its two strategies. Wang et al. [23] proposed a spectral–spatial cross-attention network to effectively exploit the spatial and spectral information in HSIs.

Inspired by Transformer [19], Peng et al. [24] proposed a spatial–spectral transformer with cross-attention, which extracts interactive spatial–spectral fusion features using a weighted shared cross-attention module and captures robust semantic features through a spatial–spectral weighted sharing mechanism. Ni et al. [25] proposed a selective spectral–spatial aggregation Transformer network (S2ATNet) to address the challenges of spectral–spatial interactive feature representation across cross-modal data. Xu et al. [26] proposed the

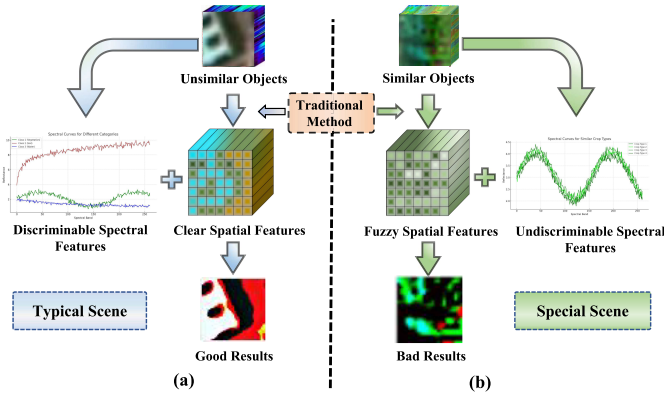

Fig. 1. (a) In typical hyperspectral scenarios, land-cover categories with significant class differences exhibit distinct differences in both spatial and spectral features, allowing traditional methods to effectively capture these distinctions and achieve ideal classification outcomes. (b) In special scenarios, similar categories display subtle spectral and spatial differences, which traditional methods struggle to capture at a fine-grained level, leading to less-than-ideal classification results.

cross-spatial–spectral dense Transformer (CS2DT) for spatial–spectral feature extraction and feature fusion. The 2-D or 3-D convolutions are commonly utilized as fundamental operations for extracting spatial or spectral features. However, due to the sensitivity of these convolutions to image rotation, they often lack robustness to rotations in HSIs. Yang et al. [27] proposed a cross-attention spectral–spatial network (CASSN) to alleviate the problem of HSI rotation. However, existing deep learning-based HSI classification methods often neglect noise and blur degradation factors during training, leading to suboptimal performance in datasets contaminated with noise and blur. Wang et al. [28] proposed a variational optimization-guided two-stage network (VOGTNet) to address performance challenges in multispectral and panchromatic image fusion with noise and blur. What is more, to address the performance and efficiency challenges arising from high-dimensional data redundancy in HSI classification using selective state-space architectures, He et al. [29] proposed a lightweight interval group spatial–spectral Mamba framework. This framework enables the extraction of multidirectional and multiscale global spatial–spectral information through grouped and hierarchical mechanisms.

With the development of drones, satellite remote sensing, and deep learning, mineral detection is increasingly shifting toward data-driven decision-making. However, compared with typical scenes, the HSI classification task in mineral scenarios is complicated by the high similarity of spectral and spatial information, making it difficult to extract useful features from a large number of spectral bands and accurately classify different mineral targets. In typical hyperspectral scenarios, different land-cover categories generally exhibit significant distinguishability in both spatial and spectral features. As shown in Fig. 1(a), land covers with large category differences (e.g., water, vegetation, and soil) exhibit obvious spatial differences, which can be easily captured by traditional attention-based methods that focus on spatial regions strongly correlated with the classification result. Additionally, the noticeable spectral differences further distinguish different

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categories, leading to ideal classification outcomes. However, in special scenes, as shown in Fig. 1(b), the spectral and spatial differences between mines are often subtle, especially between similar categories (e.g., different mine species). These mines tend to share highly similar spatial and spectral characteristics, which makes it challenging for traditional methods to capture the fine distinctions between them. As a result, the classification results are often less satisfactory. Therefore, there are absolutely few current methods [30], [31] for similar objects HSI classification tasks, and these methods do not effectively solve the problems we mentioned.

To address the challenge of distinguishing spatial and spectral features in mineral scenes, we propose an HSI classification network based on spatial–spectral perception (SSPNet). The SSPNet separates feature extraction into two stages: spatial perception and spectral perception, directly targeting the high similarity of mineral features. In the spatial perception stage, we use Fourier transforms to remap difficult-to-differentiate 3-D spatial features into subspaces of different frequency domains. Specifically, the Fourier transform decomposes the 3-D spatial cube into low-frequency components (representing macroscopic structures, e.g., vein trends and distribution ranges) and high-frequency components (representing microscopic textures, e.g., grain boundaries and internal fractures). For similar mineral species, macroscopic structures (e.g., vein shapes) often overlap, but microscopic textures (e.g., grain size distributions) differ subtly. To amplify these hidden differences, we apply our proposed triple-cross-attention (TCA) to extract features from these subspaces. The TCA mechanism models self-correlation in 3-D spatial features by implementing multidirectional pooling compression on high-frequency (via small pooling kernels) and low-frequency (via large pooling kernels) features, respectively. It fuses these compressed features to generate attention weights across dimensions, followed by cross-directional weighting. The design not only efficiently captures fine-grained spatial distinctions between similar mineral species but also significantly reduces the computational complexity of 3-D feature processing. In the spectral perception stage, due to the high dimensionality of spatial–spectral features, we design a spectral linear perceptron (SLP) based on absolute linear attention (ALA). To enhance the influence of nearby spectral information on the central spectral values, we introduce absolute positional information into the attention mechanism. It enables the model to focus on subtle spectral differences that distinguish similar mineral species. This design strengthens the internal spectral correlation modeling and reduces interference from irrelevant bands.

In summary, the contributions of this work can be summarized as follows.

1) We designed a spatial–frequency perceptron (SFP) to distinguish spatial features that are difficult to identify between similar categories. This module first remaps indistinguishable spatial features to high and low frequency space by the Fourier transform and then we use the proposed TCA to effectively extract different spatial–frequency features, respectively, while ensuring less computational complexity.
2) We present an SLP based on ALA used to accurately model spectral correlations between similar categories. This module not only implements linear self-attention based on the interaction between spectral features at different scales, but also introduces the idea of absolute positional weighting (APW) in HSIs to enhance the positive effects between close spectra.
3) Our method achieves superior and stable classification performance on three publicly available and challenging mineral HSI classification datasets. Additionally, it also achieves optimal results on an HSI classification dataset from an agricultural scenario, further validating the robustness of our proposed method.

The structure of this article is arranged as follows. Section II introduces the related works about our method. Section III outlines our proposed approach. Section IV presents the experiments conducted to validate our proposed method. Finally, Section V concludes this article and suggests future research directions.

II. Related Work

In this section, we provide an overview of the most relevant research developments categorized into three major areas: frequency-domain decomposition, nonlocal perception, linear attention mechanisms, and position encoding with attention. Our motivation for designing our modules is also analyzed for our task to better understand our approach.

A. Frequency-Domain Decomposition

High-similarity mineral HSI data complicate the extraction of discriminative spatial features due to overlapping frequency components. Traditional spatial feature extraction techniques struggle to separate high-frequency texture details from low-frequency structural information effectively. To address these challenges, Kavalerov et al. [32] proposed the 3-D Fourier scattering transform, which combines frequency-domain decomposition through time–frequency representations with neural network architectures, enabling effective capture of spectral content and enhancing spectral–spatial classification of hyperspectral imagery. Building on this, Cheng et al. [33] introduced a dual-branch deep stochastic adaptive Fourier decomposition network (DSAFDNet), which also leverages frequency-domain decomposition to improve HIS classification, particularly by requiring fewer labeled samples and reducing training time. In a similar vein, Chen et al. [30] proposed a fractional Fourier-based frequency-spatial–spectral prototype network (FrFSSPN), incorporating frequency-domain decomposition through fractional Fourier transform (FrFT) to enhance HSI classification, addressing open-set challenges and improving class differentiation. Also, Ni et al. [34] proposed a frequency-domain-based network (FDNet) to address the problem of efficiently capturing spatial, spectral, and elevation information while handling data redundancy and enhancing feature representation of land cover in cross-modal data.

The above methods all utilize frequency-domain decomposition techniques to enhance HSI classification performance.

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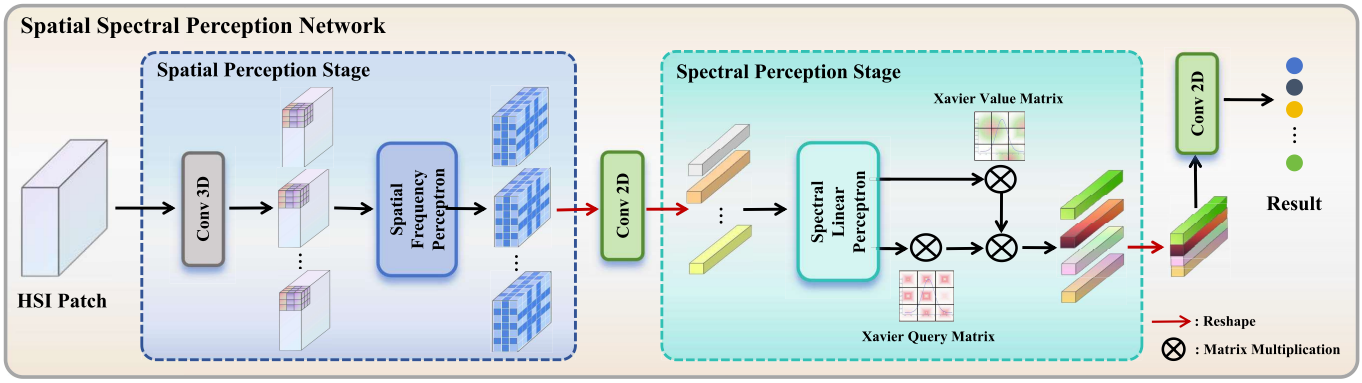

Fig. 2. Structure of the proposed SSPNet. The SSPNet proposed for mineral HSI classification consists of two phases, spatial and spectral sensing, aiming to accurately identify and distinguish similar features from spatial and spectral data.

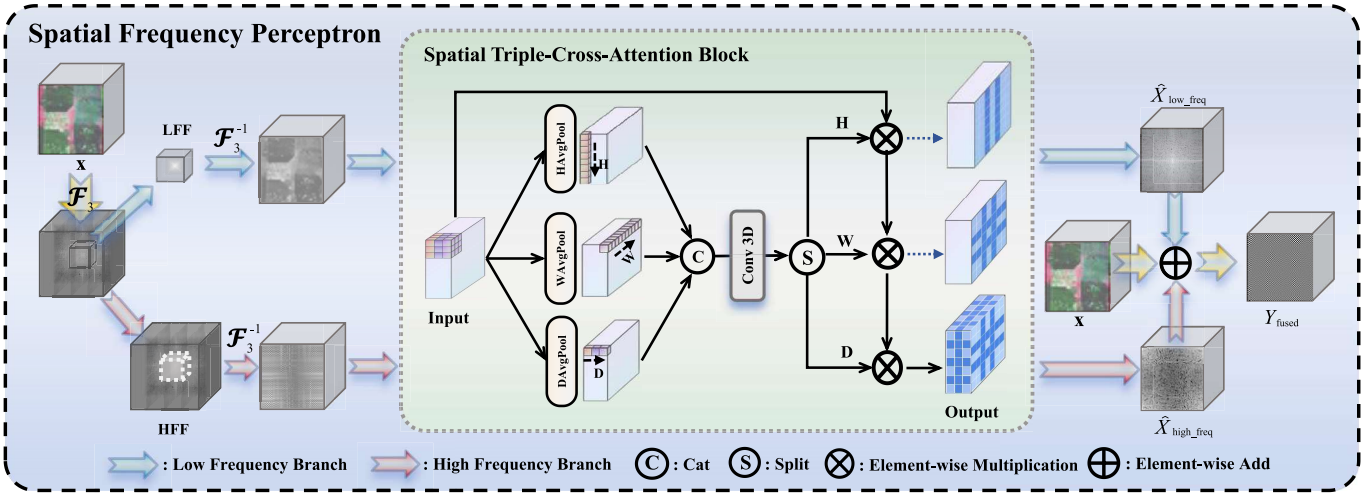

Fig. 3. SFP module. The SFP uses the Fourier transform to separate spatial features into high- and low-frequency components and employs TCA to capture multilevel spatial information by correlating different positions in a 3-D space.

However, these approaches primarily focus on feature separation in general scenarios and do not design specialized modules for the problem of highly overlapping spatial features among similar classes in mineral HSIs. The proposed SFP explicitly applies the Fourier transform to remap mineral spatial features, decomposing indistinguishable 3-D spatial features into low-frequency (macroscopic structures, e.g., vein trends) and high-frequency (microscopic textures, e.g., grain boundaries) subspaces. By combining with TCA to extract features from different frequency domains, SFP effectively addresses the insufficient separation of high-frequency texture details and low-frequency structural information in mineral scenarios, significantly improving the discriminative ability of spatial features for similar minerals.

B. Nonlocal Perception

Traditional convolution-based models are often confined to local feature extraction, overlooking the long-range spatial–spectral dependencies inherent in hyperspectral datasets. Effectively modeling these relationships is crucial for distinguishing subtle differences in mineral scenarios, where fine-grained feature differentiation is often necessary.

To address this limitation, Wang et al. [35] introduced nonlocal operations as a versatile building block for capturing long-range dependencies in computer vision tasks. This approach not only achieved state-of-the-art performance in video classification but also enhanced static image recognition tasks such as object detection and pose estimation. Building on this foundation, Zhang et al. [36] further improved nonlocal operations by proposing a residual nonlocal attention network (RNLAN) for high-quality image restoration. This network addressed the shortcomings of prior methods that treated spatial and channel-wise features uniformly and relied on local convolutions. Similarly, Wang et al. [37] introduced a pyramid dual nonlocal attention network for exemplar-based colorization, effectively capturing long-range dependencies and multiscale correlations, thus achieving photo-realistic colorization with full semantic consistency between target grayscale and reference color images. Furthermore, Chen et al. [38] presented an end-to-end learned lossy image compression approach, leveraging nonlocal attention optimization and improved context (NLAIC) modeling to enhance compression efficiency. By exploiting both global and local correlations, adaptive bit allocation, and improved entropy modeling, their method achieved state-of-the-art performance across multiple datasets. Meanwhile, Mei et al. [39] proposed a novel nonlocal sparse attention (NLSA) mechanism for single image super-resolution (SISR), combining the long-range modeling

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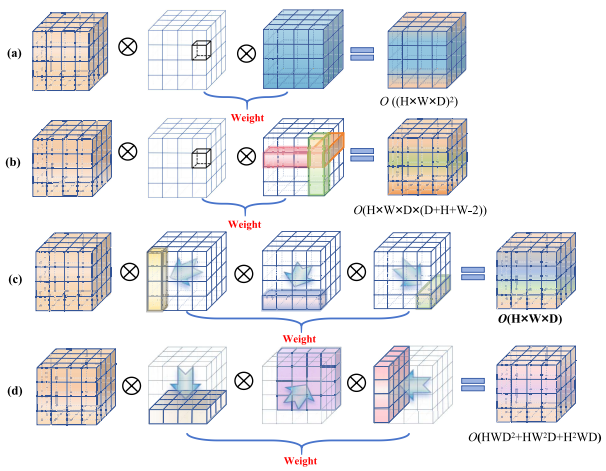

Fig. 4. Comparison of nonlocal feature-extracted structures in different 3-D spaces. (a) Nonlocal block [35]. (b) Criss-cross-3D attention [47]. (c) Proposed TCA_low (TCA_Low). (d) Proposed TCA_high (TCA_High).

capabilities of nonlocal operations with the efficiency of sparse representations.

Traditional nonlocal perception methods can capture long-range spatial–spectral dependencies but suffer from high computational complexity when processing 3-D hyperspectral data, making them ill-suited for efficient handling of large-scale hyperspectral datasets. The proposed TCA reduces computational complexity by designing small/large pooling kernels for high-frequency and low-frequency features, respectively. While maintaining the capability to model long-range dependencies, TCA significantly enhances the efficiency of 3-D spatial feature extraction, making it more suitable for fine-grained analysis of large mineral hyperspectral data.

C. Linear Attention Mechanisms

High-dimensional spectral data in HSIs lead to inefficiencies in standard self-attention mechanisms, whose computational complexity scales quadratically with the input size. This limitation hampers their application to large-scale hyperspectral datasets. To address such challenges, Bai et al. [40] proposed the linear programming incremental learning classifier (LPILC), which enhances HSI classification by enabling deep learning models to rapidly adapt to new data classes with minimal computational resources. The method effectively leverages linear attention to maintain high performance on both original and new classes, requiring only one shot of new class data. Similarly, Zhao and Li [41] proposed the deep spectral clustering with regularized linear embedding (DSCRLE), which enhances HSI segmentation by incorporating linear attention. This improves scalability and generalization, effectively combining deep learning with spectral graph theory for more efficient and accurate unsupervised clustering. In the context of hyperspectral unmixing, Hong et al. [42] introduced the augmented linear mixing model (LMM), which addresses spectral variability by incorporating a data-driven learning strategy. Linear attention is employed to learn spectral variability dictionaries, thereby enhancing the estimation of abundance maps in the presence of environmental and instrumental effects.

Existing linear attention methods reduce complexity through linear computations but fail to fully consider the critical role of absolute spectral positional information in distinguishing similar minerals. The proposed SLP integrates APW into linear attention (ALA). By dynamically enhancing spectral responses in central regions and suppressing edge noise via a learnable positional weight matrix, SLP strengthens the influence of nearby spectra on the central spectrum, addressing the limitation of traditional linear attention in underutilizing positional information for similar spectral modeling. This makes SLP better suited for fine-grained discrimination of similar spectra in mineral hyperspectral data.

D. Position Encoding

Position-aware mechanisms play a crucial role in HSI classification, as they enable the effective integration of spatial and spectral information, which is essential for accurately capturing the complex features of hyperspectral data. Traditional models often struggle with properly fusing these 2-D, leading to suboptimal performance. However, approaches like the HSI Transformer [43], SSFormer [44], and DSPCANet [45] demonstrate how positional encoding can overcome these limitations. By incorporating dynamic spatial sampling, conditional positional encoding, and channel attention, these models improve the ability to preserve fine details and capture multiscale features, which are essential for HSI classification tasks. Moreover, the SSTE-Former [46] introduces hashed positional embeddings that further enhance the model's capability to process complex spatial–spectral interactions, leading to improved classification accuracy and segmentation precision.

Existing position encoding methods fuse spatial–spectral information through dynamic sampling or hashing encoding, but often rely on static or conditional encoding, failing to optimize for the spatial heterogeneity of mineral distributions. The proposed APW dynamically modulates the weights of spatial positions on spectral features via a learnable absolute positional weight matrix. This effectively addresses the spatial heterogeneity of mineral distributions and spectral mixing, and compared to traditional position encoding, APW better highlights the local spatial aggregation features of similar minerals, improving the specificity of feature representation and classification performance.

III. Proposed Method

Fig. 2 shows the proposed spatial–spectral perception network used for HSI classification, which mainly includes two stages: the spatial perception stage and the spectral perception stage. The two substages accurately identify and distinguish similar features derived from space and spectra, respectively.

A. Spatial–Frequency Perceptron

In mineral scenes, it is often challenging to visually distinguish between mines with similar characteristics using spatial texture features. Therefore, before extracting spatial features, we remap the spatial features, which are difficult to differentiate between similar mines, into high- and low-frequency feature subspaces using the Fourier transform. The

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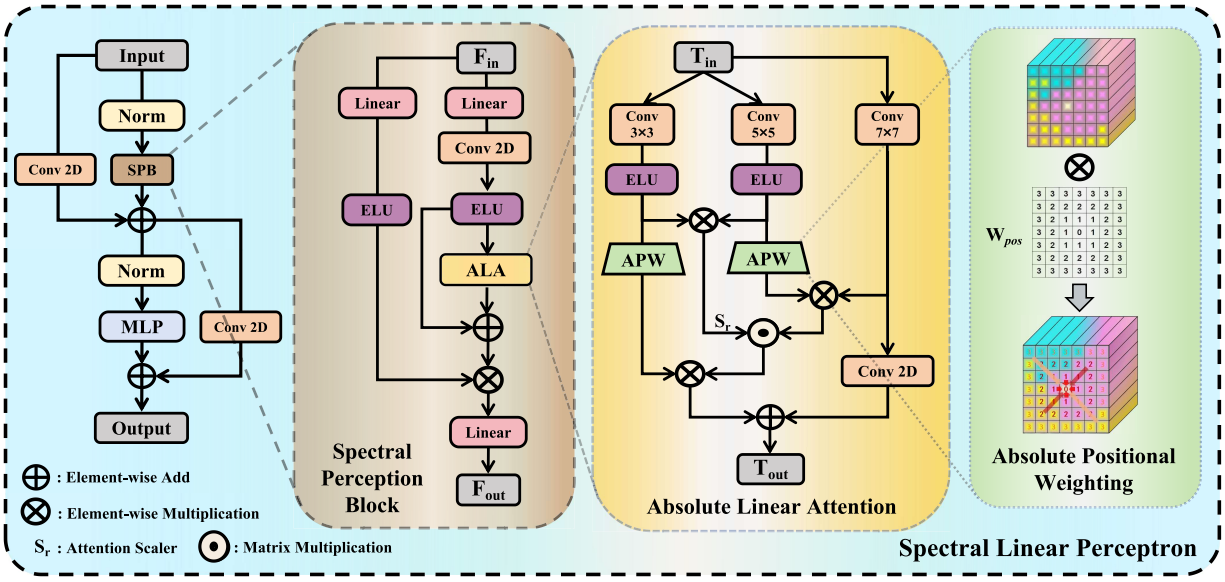

Fig. 5. SLP leverages a dual-branch architecture (integrating the SPB and deep convolution) to extract global spectral and local spatial features, thereby enhancing cross-modal representation. In this framework, the SPB incorporates ALA to capture spectral positional dependencies; ALA, in turn, strengthens global modeling through multiscale convolutions combined with APW. Finally, APW employs a learnable positional weight matrix to dynamically enhance spectral responses in central regions while suppressing edge noise.

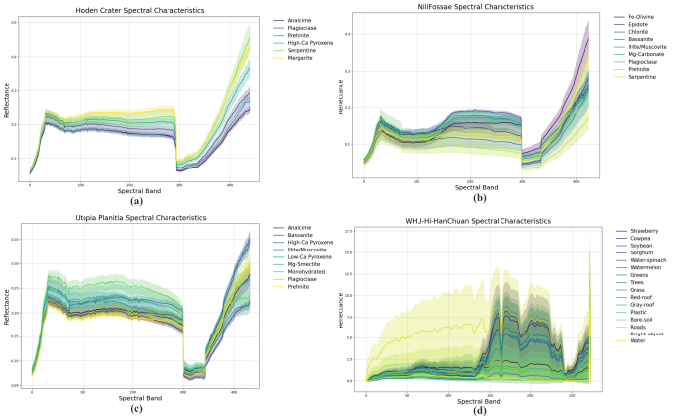

Fig. 6. Spectral characteristics on the four HSI classification datasets. (a) Holden Crater. (b) Nili Fossae. (c) Utopia Planitia. (d) WHU-Hi-HanChuan.

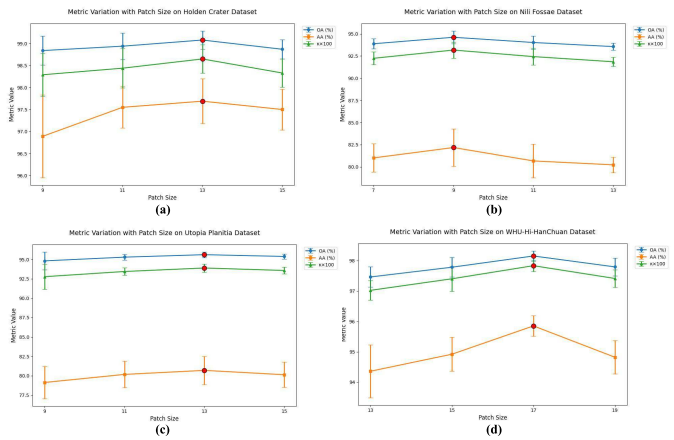

Fig. 7. Classification results of different patch sizes on the four datasets. (a) Holden Crater. (b) Nili Fossae. (c) Utopia Planitia. (d) WHU-Hi-HanChuan.

low-frequency component represents the overall structure of the image (such as general outlines and background), while the high-frequency component captures detailed information (such as textures, edges, etc.) Additionally, to better capture global spatial features, we design an efficient TCA to establish correlations between different positions in a 3-D space, thus enabling the model to sense multilevel information from both high and low-frequency spaces. The process of spatial–frequency perception is shown in Fig. 3.

1) Fourier Transform: We apply the Fourier transform to the 3-D information of a spatial cube and shift the zero-frequency space to the center of the 3-D domain, resulting in a new 3-D representation. The core advantage of the Fourier transform in mineral frequency-domain decomposition lies in its global frequency resolution capability and physical interpretability: through the linear combination of sin/cos basis functions, it can directly isolate periodic structures in mineral spectra (e.g., crystal arrangements or lamellar features) and decomposes the signal into distinct high/low-frequency components via efficient computation (FFT). Specifically, low frequencies correspond to macroscopic mineral distributions (e.g., ore veins), while high frequencies correspond to microscopic details (e.g., grain boundaries). Additionally, the frequency-domain amplitude is insensitive to translation/rotation, making it well-suited for addressing the spatial–spectral coupling characteristics of hyperspectral data. In contrast, although wavelet transforms enable localized analysis, they are computationally complex; Gabor transforms, on the other hand, involve cumbersome parameter tuning. The Fourier transform thus achieves an optimal balance between global representation efficiency and the requirements for decoupling mineral features.

In detail, the spatial region within a five-pixel distance from the zero-frequency center is defined as the low-frequency region, while regions beyond five pixels are classified as

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high-frequency regions. After reclassifying the low- and high-frequency regions, the inverse Fourier transform is used to map the 3-D space block back to its original spatial distribution.

Let the input data be $\mathbf{x} \in \mathbb{R}^{C\times D\times H\times W}$, where C is the number of channels and D, H, W are the depth, height, and width of the input, respectively.

First, we apply the 3-D Fourier transform to the input $\mathbf{x}$ as

$$\mathbf{X}_{\text{fft}} = \mathcal{F}_3(\mathbf{x}) \tag{1}$$

where $\mathcal{F}_3$ denotes the 3-D Fourier transform of $\mathbf{x}$.

Next, we perform the frequency spectrum shift to move the low-frequency components to the center of the spectrum, as

$$\mathbf{X}_{\text{fft}} = \text{fftshift}(\mathbf{X}_{\text{fft}}). \tag{2}$$

The low- and high-frequency components are separated using mask operations, as

$$\mathbf{LFF} = \mathbf{X}_{\text{fft}} \times \mathbf{M}_{\text{low}} \tag{3}$$

$$\mathbf{HFF} = \mathbf{X}_{\text{fft}} \times \mathbf{M}_{\text{high}} \tag{4}$$

where $\mathbf{M}_{\text{low}}$ and $\mathbf{M}_{\text{high}}$ are the low- and high-frequency masks, respectively, used to filter the corresponding frequency components.

Then, the inverse Fourier transform is applied to map the low- and high-frequency components back to the spatial domain, as follows:

$$\mathbf{X}_{\text{low}} = \mathcal{F}_3^{-1}(\mathbf{LFF}) \tag{5}$$

$$\mathbf{X}_{\text{high}} = \mathcal{F}_3^{-1}(\mathbf{HFF}). \tag{6}$$

The final output is the spatial-domain representation of the low- and high-frequency components.

The advantage of the Fourier transform lies in its ability to effectively separate information of different frequencies in the frequency domain. This separation allows the model to independently process low-frequency and high-frequency information based on the task's requirements, resulting in more flexible and efficient utilization of information.

2) Triple-Cross-Attention: After obtaining the 3-D features from different frequency subspaces, we design a TCA for 3-D spatial feature extraction that captures the self-correlation between different positions. First, we apply 3-D adaptive average pooling in the three directions (height, width, and depth) to compress features. Then, the pooling results from different directions are concatenated and passed through a $1\times 1\times 1$ convolution layer. Next, the resulting tensor is reshaped back to its original dimensions, and after passing through a sigmoid activation, we obtain the attention weights for different spatial dimensions. Finally, the original input is multiplied by these attention weights along each spatial dimension to capture the correlation information between different positions in different directions.

In HSI analysis, the synergistic extraction of high-frequency features (e.g., local textures, subtle spectral variations, and spatial edges) and low-frequency features (e.g., global spectral trends, regional homogeneity, and structural backgrounds) proves critical for classification and detection tasks. As illustrated in Fig. 4(c), the method employing aggressive dimensional compression pooling effectively models long-range spectral dependencies across channels, making it particularly suitable for extracting large-scale land-cover features dominated by low-frequency signatures. In addition, the approach in Fig. 4(d) achieves refined spectral–spatial interactions through 3-D tensor contraction operations. Its dimension-preserving pooling strategy optimally captures high-frequency details in the joint spectral–spatial domain, enabling precise localization of fine-grained boundaries of small targets. At the same time, the two spatial feature extraction strategies designed by us are significantly lower in computational complexity than the (a) and (b) methods compared in Fig. 4, so they are more suitable for dealing with large data features such as HSIs. Overall, the calculation process of the TCA module is as follows.

Let the input feature be $X \in \mathbb{R}^{C\times D\times H\times W}$, where C is the number of channels and D, H, W are the depth, height, and width dimensions of the feature, respectively.

For low- and high-frequency features, different pooling operations are applied.

1) For high-frequency features, small pooling kernels P_d, P_h, P_w are used as follows:

$$X_d = P_d(X), \quad X_h = P_h(X), \quad X_w = P_w(X). \tag{7}$$

2) For low-frequency features, large pooling kernels p_d, p_h, p_w are used as follows:

$$X_d = p_d(X), \quad X_h = p_h(X), \quad X_w = p_w(X). \tag{8}$$

After pooling, the features are concatenated and passed through a $1\times 1\times 1$ convolution to fuse the features, as shown in the following equation:

$$X_{\text{fusion}} = \text{Conv3D}(X_d \parallel X_h \parallel X_w). \tag{9}$$

Then, the resulting feature map X_{fusion} is split back into X_d, X_h, X_w.

The weighting process involves applying the sigmoid activation function, as follows:

$$A_d = \sigma(X_d), \quad A_h = \sigma(X_h), \quad A_w = \sigma(X_w) \tag{10}$$

where σ denotes the Sigmoid activation function.

Finally, the output is computed as

$$\hat{X} = X \cdot ((A_d \cdot A_h) \cdot A_w). \tag{11}$$

The output $\hat{X}$ is the feature after applying the spatial cross-attention mechanism.

Regarding the design of the TCA, previous research has proposed various structures. We provide a simple summary of the method for constructing nonlocal spatial features, as shown in Fig. 4. While prior methods have modeled nonlocal spatial feature correlations to varying extents, traditional approaches may introduce significant computational overhead when dealing with 3-D data. For 3-D inputs, the mechanism underlying spatial weight calculation in Fig. 4(a) involves pairwise similarity computations between all spatial positions, resulting in a computational complexity of $O((H\times W\times D)^2)$. In Fig. 4(b), the spatial weight calculation principle focuses on computing similarities between a given spatial position and other pixels along three directions—within the same row, same column, and same spectral dimension (excluding the position itself). By the way,

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TABLE I

DETAILS AND PROPORTION OF SAMPLES DIVISION OF THREE DATASETS

Dataset Scene	Dataset Name	Spatial Resolution	Bands Number	Classes Number	Pixel Size	Training Proportion	Testing Proportion	Patch Size
Mars	Holden Crater	18m	440	6	418×595	1%	99%	13
Mars	Nili Fossae	18m	425	9	487×563	1%	99%	9
Mars	Utopia Planitia	18m	432	9	478×595	1%	99%	13
Agriculture	WHU-Hi-HanChuan	0.109m	274	16	1217×303	1%	99%	17

TABLE II

DETAILS AND NUMBER OF SAMPLES FOR DIFFERENT DATASETS

Class No.	Holden Crater			Nili Fossae			Utopia Planitia			WHU-Hi-HanChuan		
	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing
1	Analcime	9	931	Fe-Olivine	85	8478	Analcime	17	1734	Strawberry	456	44279
2	Plagioclase	15	1457	Epidote	10	986	Bassanite	58	5749	Cowpea	231	22522
3	Prehnite	15	1545	Chlorite	4	382	High-Ca Pyroxene	4	428	Soybean	105	10182
4	High-Ca Pyroxene	69	6894	Bassanite	33	3253	Illite/Muscovite	3	269	Sorghum	54	5299
5	Serpentine	87	8610	Illite/Muscovite	2	226	Low-Ca Pyroxene	68	6706	Water spinach	12	1188
6	Margarite	5	453	Mg-Carbonate	49	4818	Mg-Smectite	12	1154	Watermelon	46	4487
7	-	-	-	Plagioclase	10	1047	Monohydrated sulfate	3	337	Greens	59	5844
8	-	-	-	Prehnite	12	1149	Plagioclase	2	200	Trees	183	17795
9	-	-	-	Serpentine	62	6104	Prehnite	6	588	Grass	95	9374
10	-	-	-	-	-	-	-	-	-	Red roof	106	10410
11	-	-	-	-	-	-	-	-	-	Gray roof	174	16737
12	-	-	-	-	-	-	-	-	-	Plastic	39	3640
13	-	-	-	-	-	-	-	-	-	Bare soil	91	9025
14	-	-	-	-	-	-	-	-	-	Road	188	18372
15	-	-	-	-	-	-	-	-	-	Bright object	11	1125
16	-	-	-	-	-	-	-	-	-	Water	768	74633
Total test/train	200/19890			267/26443			173/17165			2618/254912		

TABLE III

COMPARISON RESULTS OF DIFFERENT SIZES IN FREQUENCY SEPARATION ON NILI FOSSAE

Metric	The size of frequency separation					
	3	4	5	6	7	8
OA (%)	94.16±0.51	94.56±0.39	**94.61±0.73**	94.23±0.78	94.09±0.63	92.96±0.92
AA (%)	80.74±1.99	81.87±1.29	**82.18±2.10**	81.54±1.88	81.74±2.33	78.99±2.24
κ*100	92.60±0.64	93.12±0.49	**93.17±0.92**	92.69±0.99	92.52±0.80	91.09±1.16

TABLE IV

COMPARISON RESULTS OF DIFFERENT 3-D SPATIAL ATTENTION ON NILI FOSSAE

Metric	NonLocal3D	CC3D*1	CC3D*3	TCA(Ours)
OA (%)	94.28±0.59	94.21±0.65	94.35±0.65	**94.61±0.73**
AA (%)	81.22±2.42	81.16±1.72	**82.58±0.84**	82.18±2.10
κ*100	92.76±0.74	92.77±0.82	92.86±0.82	**93.17±0.92**
Total FLOPs(G)	2.32	1.23	1.55	1.09

TABLE V

COMPARISON RESULTS OF SLP ON NILI FOSSAE

Metric	Transformer	AdaptFormer	CasViT	DyFilter	SLP(Ours)
OA (%)	93.73±1.26	94.35±0.35	94.09±0.71	94.27±0.41	**94.61±0.73**
AA (%)	81.09±2.39	81.62±1.09	81.12±2.59	81.25±1.54	**82.18±2.10**
κ*100	92.07±1.60	92.85±0.45	92.52±0.90	92.74±0.52	**93.17±0.92**

according to the original paper [47], the CC3D should be conducted three times—leading to a computational complexity of $O(3 \times (H \times W \times D) \times (H + W + D - 2))$. For our proposed pooling-based TCA_Low and TCA_High, as illustrated in Fig. 4(c) and (d), we first perform feature compression along the 3-D (row, column, and spectral) using pooling operations, then compute similarities based on the compressed features. This approach yields computational complexities of $O(H \times W \times D)$ and $O(D^2\mathrm{HW} + \mathrm{DH}^2W + \mathrm{DHW}^2)$, respectively. It is worth noting that high-frequency features, which are more focused on details, require a smaller pooling kernel compared to low-frequency features, which helps in better feature compression.

3) Feature Fusion: Due to the evident complementary nature of features between different frequency domains, we fuse the 3-D spatial correlation information obtained from

TABLE VI

COMPARISON RESULTS OF LINEAR ATTENTION ON NILI FOSSAE

Metric	Eff-Att	Agent-Att	MLLA	PolaLA	ALA(Ours)
OA (%)	94.26±0.67	94.29±0.39	94.15±0.45	94.48±0.48	**94.61±0.73**
AA (%)	80.50±2.20	81.05±1.96	81.38±1.64	82.09±2.35	**82.18±2.10**
κ*100	92.74±0.84	92.76±0.49	92.59±0.57	93.01±0.62	**93.17±0.92**

different frequency spaces and perform a residual connection with the original input. The process of fusion can be expressed as

$$Y_{\text{fused}} = \hat{X}_{\text{low_freq}} + \hat{X}_{\text{high_freq}} + \mathbf{x} \tag{12}$$

where $\hat{X}_{\text{low_freq}}$ and $\hat{X}_{\text{high_freq}}$ are the low- and high-frequency features after processing through TCA_Low and TCA_High, and x represents the input feature.

The Fourier transform enables the model to analyze and process the image from the frequency domain, capturing features at different frequencies. By combining this with a 3-D TCA, the model can enhance its learning of spatial information at the frequency-domain level, thereby improving the modeling of both local and global features in the image.

B. Spectral Linear Perceptrons

Spectral data typically refers to data acquired through sensors that capture multiple wavelength channels. In HSIs, neighboring pixels of the same object often exhibit strong spatial correlation. By utilizing a self-attention mechanism, models can compute the spatial similarity between different pixels and assign appropriate weights to them. Additionally, there may be strong correlations or complementary relationships between different spectral bands. Some bands may contain features specific to a material, while others may provide background or irrelevant information. The attention mechanism helps the model dynamically adjust the weight of each spectral band based on its contribution, thereby enhancing classification accuracy. To this end, we propose an SLP that

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Algorithm 1 Absolute Linear Attention

Input: $\mathbf{T}_{\text{in}} \in \mathbb{R}^{B\times N\times C}$($B$ batch size, N spatial tokens, C spectral dimension)
Output: Enhanced features $\mathbf{T}_{\text{out}} \in \mathbb{R}^{B\times N\times C}$
Initialize: $H \leftarrow$ num_heads, $d \leftarrow$ head_dim $\leftarrow C/\text{H}$, $\epsilon \leftarrow 10^{-6}$
Nonlinear Transformation:
$\mathbf{Q}, \mathbf{K}, \mathbf{V} \leftarrow Conv_3(\mathbf{T}_{\text{in}}), Conv_5(\mathbf{T}_{\text{in}}), Conv_7(\mathbf{T}_{\text{in}})$
$\mathbf{Q}' \leftarrow \text{ELU}(\mathbf{Q}) + 1.0$
$\mathbf{K}' \leftarrow \text{ELU}(\mathbf{K}) + 1.0$
Absolute Positional Weighting:
$\mathbf{Q}_{\text{APW}} \leftarrow \text{APW}(\texttt{reshape}(\mathbf{Q}'))$
$\mathbf{K}_{\text{APW}} \leftarrow \text{APW}(\texttt{reshape}(\mathbf{K}'))$
Multihead Processing:
for $h \leftarrow 1$ H **do**
  Decompose $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h$ from $\mathbf{Q}_{\text{APW}}, \mathbf{K}_{\text{APW}}, \mathbf{V}$
  Compute normalization factor:
  $\mathcal{Z}_h \leftarrow 1/(\mathbf{Q}_h\mathbf{K}_h^\top + \epsilon)$
  Compute projected features:
  $\mathbf{KV}_h \leftarrow (\mathbf{K}_h^\top \cdot N^{-0.5})(\mathbf{V}_h \cdot N^{-0.5})$
  $\mathbf{Y}_h \leftarrow \mathbf{Q}_h\mathbf{KV}_h \odot \mathcal{Z}_h$
end for
Feature Fusion:
$\mathbf{Y} \leftarrow \texttt{concat}(\{\mathbf{Y}_h\}_{h=1}^{H}) \in \mathbb{R}^{B\times N\times C}$
$\mathbf{v}' \leftarrow \text{Conv2D}(\texttt{reshape}(\mathbf{V}))$
$\mathbf{T}_{\text{out}} \leftarrow \mathbf{Y} + \texttt{reshape}(\mathbf{v}')$
Return $\mathbf{T}_{\text{out}}$

combines learned absolute positional weighting with a computationally efficient spectral-based linear attention mechanism to jointly model spatial and spectral correlations. The structure of the SLP is shown in Fig. 5. The proposed SLP innovatively achieves the decoupled learning of spatial–spectral features through a dual-branch structure. This module utilizes a spectral perception block (SPB) and deep convolution to parallelly extract global spectral features and local spatial features and constructs a hybrid optimization path through dual residual connections, further enhancing the cross-modal representation capability of spatial–spectral information. The overall algorithm process of SLP can be summarized as

$$\mathbf{X}' = \text{SPB}\left(\text{Norm}\left(\mathbf{X}_{\text{Input}}\right)\right) + \text{dc}\left(\mathbf{X}_{\text{Input}}\right) \tag{13}$$

$$\mathbf{X}_{\text{Output}} = \text{dc}\left(\mathbf{X}'\right) + \left(MLP\left(\text{Norm}\left(\mathbf{X}'\right)\right)\right) \tag{14}$$

where $\mathbf{X}_{\text{Input}}$ represents the input feature, dc represents depthwise convolution, Norm represents layer normalization, and SPB represents spectral perception block.

1) Spectral Perception Block: The SPB innovatively constructs a hybrid attention mechanism. It achieves local spectral feature compression through the cascade of depthwise separable convolutions and linear projections, while leveraging ALA to capture positional dependencies between spectra and overlay residuals to preserve details. Finally, dynamic calibration of the weight distribution between the original input features and deep features is performed through cross-layer matrix multiplication, achieving an optimal balance between computational complexity and spectral sensitivity. The overall

Algorithm 2 Absolute Positional Weighting

Require: Input feature map $x \in \mathbb{R}^{B\times H\times W\times C}$
Ensure: Weighted feature map $x' \in \mathbb{R}^{B\times H\times W\times C}$
Calculate center coordinates: $c_h \leftarrow \lfloor \frac{H-1}{2} \rfloor$, $c_w \leftarrow \lfloor \frac{W-1}{2} \rfloor$
Generate position indices: $\Delta h_{ij} \leftarrow i - c_h$, $\Delta w_{ij} \leftarrow j - c_w \quad \forall i, j$
Compute max offsets: $D_h \leftarrow \max(\|\Delta h\|_\infty, c_h)$, $D_w \leftarrow \max(\|\Delta w\|_\infty, c_w)$
Initialize learnable weights: $\mathbf{W} \in \mathbb{R}^{(2D_h+1)\times(2D_w+1)\times C}$
Normalize indices: $\mathcal{I}_h \leftarrow \Delta h + D_h$, $\mathcal{I}_w \leftarrow \Delta w + D_w$
Retrieve spatial weights: $\mathbf{W}_{pos} \leftarrow Sigmoid(\mathbf{W}[\mathcal{I}_h, \mathcal{I}_w, :])$
$x' \leftarrow x \odot \mathbf{W}_{pos}$

algorithm process of SPB can be summarized as

$$\mathbf{F}' = \sigma\left(\left(\text{DWConv}\left(\text{Proj}\left(\mathbf{F}_{\text{in}}\right)\right)\right)\right) \tag{15}$$

where $\mathbf{F}_{\text{in}}$ represents the input feature, DWConv represents depth-wise separable convolution, σ represents ELU activation function, and Proj represents linear layer

$$\mathbf{F}_{\text{out}} = \text{Proj}\left(\left(\text{ALA}\left(\mathbf{F}'\right) + \mathbf{F}'\right) \odot \sigma\left(\text{Proj}\left(\mathbf{F}_{\text{in}}\right)\right)\right) \tag{16}$$

where ALA represents absolute linear attention and $\odot$ represents matrix multiplication.

2) Absolute Linear Attention: Linear attention efficiently captures spectral feature interactions by dynamically weighting spectral bands per pixel, replacing traditional dot-product computations with key averaging and scaling factors. This design reduces complexity from $O(N^2)$ to $O(N)$, significantly lowering computational overhead for hyperspectral data. Building on this, ALA further enhances global modeling through multiscale convolutions that generate query/key/value matrices, integrating localized spatial patterns via ELU activation and APW. A hybrid gating mechanism combines attention and convolution, preserving long-range dependencies while refining local details, thereby harmonizing computational efficiency with precise spectral–spatial sensitivity. The algorithm flow of ALA is shown in Algorithm 1.

In high-dimensional data, the ALA effectively integrates data from different dimensions (such as spatial and spectral) into a standardized feature representation. The attention mechanism further establishes cross-domain relationships between HSI data, forming more powerful spatial–spectral feature representations.

3) Absolute Positional Weighting: In the task of mineral HSI classification, mineral samples often exhibit specific spatial aggregation patterns in the image. By utilizing a learnable absolute position weight matrix, the absolute positional weighting can dynamically enhance the spectral response of the central region while suppressing noise interference from edge areas. It effectively addresses the challenges of spatial heterogeneity in mineral distribution and spectral mixing through a collaborative mechanism of spatial location-based weighting modulation and spectral feature adaptive enhancement. The detailed flow of the algorithm is shown in Algorithm 2.

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TABLE VII
ABLATION RESULTS OF FOURIER TRANSFORM ON THE THREE DATASETS

Metric	Holden Crater		Nili Fossae		Utopia Planitia	
	No_Fourier	Ours	No_Fourier	Ours	No_Fourier	Ours
OA (%)	98.74±0.23	**99.08±0.21**	94.37±0.53	**94.61±0.73**	94.85±0.43	**95.61±0.37**
AA (%)	97.37±0.36	**97.69±0.51**	81.80±1.04	**82.18±2.10**	78.95±1.86	**80.69±1.83**
κ*100	98.14±0.34	**98.65±0.32**	92.87±0.66	**93.17±0.92**	92.83±0.59	**93.89±0.52**

TABLE VIII
ABLATION RESULTS OF TCA_LOW AND TCA_HIGH ON NILI FOSSAE

Metric	No_TCA	Only_Low	TCA_Low	Only_High	TCA_High	Ours
OA (%)	93.40±1.18	94.14±0.46	94.07±0.58	94.29±0.58	94.29±0.59	**94.61±0.73**
AA (%)	79.95±1.72	82.11±1.58	80.46±1.88	81.54±1.42	81.52±1.97	**82.18±2.10**
κ*100	91.97±1.37	92.59±0.58	92.48±0.74	92.77±0.73	92.78±0.74	**93.17±0.92**

C. Feature Separation

After applying the SLP, we use two matrices initialized from a normal distribution to separate the features. Specifically, we transform the flattened spatial–spectral features into tokens, with each token representing a feature vector. These tokens contain both spectral information and spatial information. In the two normal distribution matrices we use, $\boldsymbol{M}_Q \in \mathbb{R}^{b\times k\times c}$ represents the query vectors for each token, used to compute similarity between tokens, while $\boldsymbol{M}_V \in \mathbb{R}^{b\times c\times c}$ represents the value vectors for each token, containing the feature information of that token. In detail, b represents batch size, k represents the number of learnable tokens, and c represents the number of spectral bands. The goal is to separate features with inconsistent classes based on token similarity, facilitating the weighted aggregation of tokens to generate a final representation that accurately reflects different land-cover classes. It can be summarized as

$$T = \text{Conv2}D\left[\underbrace{\left(\text{softmax}\left(tM_Q^{\top}\right)\right)^{\top}}_{\text{Token attention weights}} \odot \cdot \underbrace{\left(tM_V\right)}_{\text{Transformed features}}\right] \tag{17}$$

where the input $\boldsymbol{t} \in \mathbb{R}^{b\times n\times c}$, $\odot$ represents matrix multiplication, and the kernel size of Conv2D is $1 \times k$.

Through weighted feature extraction and adaptive feature selection, the model focuses on the most important spectral features, while tokenization breaks down the high-dimensional spatial–spectral information into manageable units, thus improving feature expressiveness and classification accuracy. This synergy not only enhances model performance but also makes it more efficient and flexible when handling complex spatial–spectral dependencies.

IV. EXPERIMENTS

To validate the effectiveness of our proposed method as well as the individual submodules, we perform extensive comparative and ablation experiments in this section. The methods we used to contrast in comparative experiments include recent relevant studies to ensure the advancement of our method. In addition, our ablation analyses were also tested on multiple datasets to demonstrate the robustness of our proposed method.

A. Datasets

We conducted comprehensive comparative experiments with state-of-the-art methods on three challenging mineral HSI classification benchmarks: Holden Crater [48], Nili Fossae [48], and Utopia Planitia [48] datasets, containing 6, 9, and 9 distinct mine species, respectively. These datasets present significant classification challenges due to the high interclass similarity in both spatial patterns and spectral signatures among different mines. To further evaluate the cross-domain robustness of our proposed method in agricultural scenarios, we extended validation to the WHU-Hi-HanChuan hyperspectral dataset [49]—a real-world agricultural benchmark comprising 16 crop categories with remarkably similar spatial–spectral characteristics. This experimental design systematically verifies our method's effectiveness in handling fine-grained discrimination tasks across both geology and agriculture domains. Additionally, the four datasets we selected all face the challenge of imbalanced class sample sizes, which rigorously tests the model's performance on minority sample classes. Details on the datasets are shown in Tables I and II and Fig. 6.

B. Experimental Setup

1) Evaluation Metrics: We comprehensively evaluated our proposed method on three metrics, which are overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ). The larger these three indicators are, the better the classification performance is.

2) Configuration: All of our experiments were performed based on the service station of Intel Xeon Silver 4210R CPU, 64GB RAM, and an NVIDIA Quadro RTX 5000 16GB GPU.

3) Experimental Details: We validated our method on 2 machine learning-based models and nine deep learning-based HSI classification models, which are: SVM [7], RF [10], PyResnet [50], ContextualNet [51], SSRN [52], SSFTT [53], MATNet [54], SS-MSDCNet [55], SPTNet [56], CSCANet [57], and MCTGCL [48]. To reduce the error as much as possible, RF and all results of DL-based networks were performed with ten experiments and calculated standard deviations. For all methods, we kept the same settings as the original papers. In addition, we set the batch size to 64 in the proposed SSPNet, the spectral dimension after PCA is 30, and the optimizer used in the network is Adam, with a learning rate of 0.0005. The training epochs for Holden Crater, Nili Fossae, Utopia Planitia, and WHU-Hi-HanChuan are 200, 200, 300, and 150, respectively.

C. Comparison Analysis

1) Patch Size: To investigate the optimal patch size settings of our proposed method across different datasets, we conducted comparative experiments on four datasets, with the experimental results illustrated in Fig. 7. For the Holden Crater, Nili Fossae, Utopia Planitia, and WHU-Hi-HanChuan datasets, the optimal classification results were achieved when the patch sizes were set to 13, 9, 13, and 17, respectively. Generally, when the number of samples per class is larger, larger patch size settings tend to yield better classification performance.

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TABLE IX

ABLATION RESULTS OF SPB, ALA, AND ABSOLUTE POSITIONAL WEIGHTING ON THE THREE DATASETS

Metric	Holden Crater				Nili Fossae				Utopia Planitia			
	No_SPB	No_ALA	No_APW	Ours	No_SPB	No_ALA	No_APW	Ours	No_SPB	No_ALA	No_APW	Ours
OA (%)	98.55±0.17	98.77±0.23	98.80±0.24	**99.08±0.21**	93.69±0.54	93.99±0.62	94.06±0.87	**94.61±0.73**	95.35±0.45	95.40±0.43	95.48±0.22	**95.61±0.37**
AA (%)	97.06±0.50	97.30±0.41	97.41±0.50	**97.69±0.51**	80.88±2.13	81.08±1.58	81.28±2.13	**82.18±2.10**	80.07±1.68	80.22±1.51	80.14±1.59	**80.69±1.83**
$\kappa * 100$	98.09±0.36	98.28±0.34	98.39±0.36	**98.65±0.32**	91.78±1.10	92.17±0.79	92.48±1.10	**93.17±0.92**	93.46±0.54	93.60±0.60	93.70±0.30	**93.89±0.52**

TABLE X

ABLATION OF CLASSIFICATION RESULTS AND RUNTIME EFFICIENCY OF SFP AND SLP ON THE NILI FOSSAE DATASET

Metric	No_SFP_SLP	No_SFP	No_SLP	SSPNet
OA (%)	90.01±0.53	94.16±0.41	93.65±0.65	**94.61±0.73**
AA (%)	76.46±0.93	80.90±1.33	79.42±2.00	**82.18±2.10**
κ*100	86.75±0.46	92.62±0.52	91.97±0.83	**93.17±0.92**
Total Params	417796	489348	417972	489524
Avg-Infer-Time(ms)	1.23	3.40	3.55	5.01
Total FLOPs(G)	0.78	1.07	0.80	1.09

TABLE XI

COMPARISON OF RUNTIME EFFICIENCY OF DIFFERENT DEEP LEARNING-BASED METHODS ON THE NILI FOSSAE DATASET

Metric	PyResNet	ContextualNet	SSRN	SSFTT	MATNet
Total Params	21845019	263945	102873	426945	1595510
Avg-Infer-Time(ms)	6.14	11.42	2.59	2.19	8.24
Total FLOPs(G)	2.12	0.88	1.40	0.84	3.96
Metric	SS-MSDCNet	SPTNet	CSCANet	MCTGCL	SSPNet
Total Params	19449074	97392	847564	238258	489524
Avg-Infer-Time(ms)	37.27	5.27	5.12	4.47	5.01
Total FLOPs(G)	84.42	1.08	0.63	2.12	1.09

2) Division of Frequency: We conducted experimental tests on the Nili Fossae dataset to explore the division of high- and low-frequency features in the SFP. The pixel range for the division was set to $[3, 4, 5, 6, 7, 8]$, with the corresponding experimental results shown in Table III. We observed that when the parameter was set to 5, our proposed method achieved the best performance on the Nili Fossae dataset. The division of high- and low-frequency spaces is closely related to the spatial resolution of the dataset itself. When the image resolution is higher, it is necessary to preserve more high-frequency texture details. Conversely, when the image resolution is lower, the model needs to focus more on the global low-frequency contours of the image.

3) Triple-Cross-Attention: To validate the effectiveness of our proposed TCA in learning feature relationships in 3-D space, we compared TCA with two popular 3-D spatial self-attention mechanisms: NonLocal3D [35] and CC3D [47]. Specifically, we replaced TCA_Low and TCA_High with these two modules while keeping other settings consistent, and the resulting comparison is shown in Table IV. The results demonstrate that our method achieves the best performance in terms of OA and κ, and the second-best result in terms of AA. We hypothesize that the suboptimal AA result may be attributed to the fact that, for the class with extremely few samples (Class No.5) in the Nili Fossae dataset, CC3D with three iterations (consistent with the original paper [47], where this module iterates three times) extracts more detailed texture information compared to our designed TCA_High. This discrepancy results in TCA_High's classification performance on this class being lower than that of CC3D with three iterations. Notably, when using a single iteration of CC3D, our proposed method demonstrates clear superiority across all evaluation metrics. Meanwhile, through comparisons of FLOPs under different conditions, we confirm that our TCA achieves lower computational consumption than CC3D*3 with three iterations, making it more suitable for HSI tasks involving larger data volumes.

4) Spectral Linear Perceptron: The SLP we propose leverages a multihead self-attention mechanism to model the correlations between different spectral bands. To further validate the effectiveness of our method compared to other multihead self-attention-based approaches, we compared SLP with several popular methods, including [19], AdaptFormer [58], CasViT [59], and DyFilter [60]. These methods were substituted for SLP in the experiments, and the comparison results are presented in Table V. The results demonstrate that our proposed SLP outperforms all other methods across all evaluation metrics.

5) Absolute Linear Attention: The ALA we designed is more effective at focusing on absolute positional information within the space compared to other linear attention mechanisms. To validate the superiority of our proposed method over other linear attention mechanisms, we conducted experimental comparisons between ALA and several existing approaches, including EfficientAttention (Eff-Att) [61], AgentAttention (AgentAtt) [62], MLLA [63], and PolaLinearAttention (PolaLA) [64]. We replaced ALA with these four different linear attention mechanisms and compared the results on the Nili Fossae dataset. As shown in Table VI, our proposed ALA achieved the best performance across all metrics, indicating that the absolute position-weighted linear attention we introduced is more suitable for visual tasks.

D. Ablation Analysis

1) Fourier Transform: To validate the importance of the Fourier transform-based high- and low-frequency spatial–frequency separation in our SFP, we conducted ablation experiments on three datasets. In these experiments, we removed the Fourier transform-based frequency separation operation from the model and compared the results with the original ones. The results in Table VII demonstrate that the classification performance using the Fourier transform consistently outperforms the structure that does not utilize it.

2) Triple-Cross-Attention: To validate the effectiveness of our proposed TCA series methods in separately extracting low- and high-frequency spatial features, we conducted extensive ablation experiments on the Nili Fossae dataset. These experiments covered the following scenarios: removing both TCA_Low and TCA_High operations (No_TCA), using only the TCA_Low operation of the low-frequency branch (Only_Low), using the dual-branch TCA_Low operation (TCA_Low), using only the TCA_High operation of the high-frequency branch (Only_High), and using the dual-branch TCA_High operation (TCA_High). We compared the

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TABLE XII

DETAILED CLASSIFICATION ACCURACY OF HOLDEN CRATER ON DIFFERENT NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	SVM	RF	PyResNet	ContextualNet	SSRN	SSFTT	MATNet	SS-MSDCNet	SPTNet	CSCANet	MCTGCL	SSPNet
1	87.97	97.53±1.38	74.51±26.30	85.79±6.00	83.11±6.88	**99.43±0.29**	96.79±2.41	98.53±1.07	99.52±0.23	98.44±0.88	99.20±1.73	99.40±0.22
2	22.03	75.86±1.55	87.08±5.05	91.56±4.75	92.69±5.70	94.02±3.72	91.57±5.10	91.29±1.03	91.51±2.20	93.14±2.24	95.14±3.40	**95.70±1.55**
3	91.84	84.19±1.23	79.83±14.76	87.71±5.28	86.77±8.56	89.75±1.63	91.28±4.16	93.05±3.46	93.84±2.31	91.13±2.03	92.53±2.59	**94.16±1.95**
4	98.49	99.07±0.19	91.23±10.49	97.08±1.95	96.32±2.36	98.39±1.13	97.90±1.52	98.50±0.36	98.39±1.04	99.09±0.55	99.56±0.95	**99.87±0.11**
5	99.98	**100.00±0.00**	98.53±1.27	98.29±1.59	98.69±0.81	99.68±0.28	99.14±0.64	99.63±0.16	99.91±0.10	99.85±0.12	99.99±0.02	99.99±0.02
6	1.77	44.30±1.92	94.33±8.50	95.28±4.79	96.90±4.09	94.02±5.20	95.33±1.92	89.09±9.86	94.27±5.26	97.24±0.90	89.16±4.09	**97.04±1.15**
OA (%)	90.32	95.30±0.15	91.40±5.54	95.81±1.23	95.45±1.41	97.91±0.81	97.35±0.73	97.82±0.32	98.15±0.60	98.29±0.22	98.63±0.43	**99.08±0.21**
AA (%)	67.01	83.49±0.39	87.59±5.93	92.62±2.44	92.41±1.69	95.88±1.72	95.34±1.11	95.01±1.72	96.24±1.21	96.48±0.48	95.93±2.04	**97.69±0.51**
κ*100	85.19	92.99±0.22	87.32±8.10	93.80±1.83	93.27±2.12	96.92±1.20	88.93±1.44	96.79±0.48	97.27±0.87	97.48±0.32	97.97±0.64	**98.65±0.32**

TABLE XIII

DETAILED CLASSIFICATION ACCURACY OF NILI FOSSAE ON DIFFERENT NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	SVM	RF	PyResNet	ContextualNet	SSRN	SSFTT	MATNet	SS-MSDCNet	SPTNet	CSCANet	MCTGCL	SSPNet
1	97.96	97.45±0.29	96.87±2.52	98.21±1.47	97.14±2.36	96.98±1.55	97.23±1.40	98.81±0.83	95.40±1.42	98.12±0.80	**99.36±0.45**	98.48±0.61
2	0.30	34.90±4.83	75.01±13.20	81.93±4.38	75.85±8.40	74.28±5.42	69.80±7.39	**84.93±4.74**	80.38±3.66	82.04±4.87	69.32±5.94	83.08±3.19
3	0.00	9.66±4.68	**78.51±16.68**	62.35±11.41	58.31±40.09	70.99±16.34	81.94±7.36	64.63±12.37	58.35±11.09	60.52±22.11	55.94±12.43	75.65±13.48
4	92.53	94.11±0.76	79.81±9.68	87.20±3.40	86.52±2.42	91.52±3.06	93.10±1.91	94.32±0.53	92.28±2.32	94.35±1.57	94.24±1.64	**96.14±1.23**
5	0.00	0.22±0.41	**58.22±24.70**	42.65±7.99	28.45±36.08	31.68±7.16	36.33±7.80	31.81±3.53	33.72±3.02	5.58±5.64	35.40±6.20	28.45±2.59
6	86.80	95.56±0.76	84.63±3.74	95.40±3.11	84.95±6.48	89.82±2.01	91.52±2.73	87.78±5.51	91.94±3.38	97.12±1.11	94.71±4.16	**97.55±1.52**
7	18.82	46.77±4.97	**88.40±4.57**	85.14±8.04	79.56±11.74	77.05±6.04	75.74±8.68	84.20±4.90	75.86±4.86	81.79±3.46	77.71±4.82	82.69±3.20
8	23.41	55.29±4.45	75.35±16.38	80.42±5.25	77.48±10.47	84.00±6.71	83.95±6.03	86.90±3.17	79.88±4.83	**88.69±1.48**	81.97±5.57	81.50±4.20
9	93.73	93.81±0.71	91.35±3.76	95.79±1.35	91.01±2.93	95.10±0.56	94.73±1.21	94.54±0.88	94.61±1.23	95.46±0.62	95.58±0.25	**96.12±0.45**
OA (%)	81.93	87.58±0.27	87.72±2.30	92.94±0.52	89.23±2.84	91.44±0.66	91.91±1.01	92.58±1.33	91.13±1.18	93.87±0.66	93.10±0.98	**94.61±0.73**
AA (%)	45.92	58.64±1.05	80.91±4.84	81.12±1.79	75.47±9.87	79.05±2.23	80.48±2.20	80.88±2.52	78.04±1.73	78.19±3.12	78.25±2.64	**82.18±2.10**
κ*100	76.54	84.07±0.36	84.41±2.91	91.07±0.65	86.32±3.60	89.19±0.83	88.93±1.44	90.58±1.70	88.77±1.49	92.24±0.84	91.25±1.25	**93.17±0.92**

TABLE XIV

DETAILED CLASSIFICATION ACCURACY OF UTOPIA PLANITIA ON DIFFERENT NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	SVM	RF	PyResNet	ContextualNet	SSRN	SSFTT	MATNet	SS-MSDCNet	SPTNet	CSCANet	MCTGCL	SSPNet
1	61.88	**98.44±0.42**	79.86±6.69	84.95±1.33	81.73±6.90	86.39±3.56	83.03±3.59	82.95±2.26	87.35±2.66	90.02±1.83	89.94±2.82	91.14±1.65
2	98.64	98.58±0.18	86.63±3.93	89.73±5.07	91.93±2.34	98.60±0.53	97.73±1.02	96.64±1.50	98.30±0.62	99.00±0.40	**99.16±0.27**	98.72±0.41
3	0.00	32.57±5.72	76.65±7.32	72.24±13.23	**76.83±9.64**	60.47±5.24	59.11±13.25	40.09±1.62	52.89±7.83	49.32±4.02	64.39±5.87	60.70±9.23
4	0.00	26.28±6.89	78.80±13.81	70.04±19.76	80.22±14.47	65.46±9.94	70.22±16.09	**83.31±7.93**	54.65±4.97	60.07±7.91	61.00±9.56	66.02±14.77
5	94.78	98.63±0.32	96.86±1.58	98.20±1.27	96.23±2.17	98.11±0.55	98.89±0.51	99.27±0.47	97.72±1.07	**99.54±0.29**	99.26±0.31	99.46±0.47
6	66.55	95.03±1.96	84.71±6.36	85.83±9.23	85.78±10.39	92.69±4.46	90.42±4.59	96.07±2.84	95.36±2.56	98.32±1.22	95.71±1.71	**99.13±0.99**
7	0.00	80.83±7.19	96.64±5.15	96.42±4.88	97.24±3.98	95.85±3.87	95.82±2.57	97.44±1.07	97.33±2.33	98.22±1.88	95.82±3.67	**99.17±1.40**
8	0.00	0.65±0.63	**68.68±16.76**	48.69±22.41	43.62±19.34	26.63±9.17	30.85±11.32	30.00±4.33	17.94±4.11	12.21±7.63	25.18±9.57	21.16±4.54
9	0.00	88.40±5.18	77.79±12.64	64.52±8.04	69.53±14.41	88.79±5.07	84.32±9.06	**96.14±3.66**	92.99±2.05	87.60±3.91	93.76±2.94	87.02±3.24
OA (%)	80.79	93.73±0.19	89.02±1.80	89.97±2.63	90.26±1.70	94.09±0.62	93.54±1.00	94.27±0.36	93.82±0.61	95.00±0.34	95.19±0.37	**95.61±0.37**
AA (%)	35.76	68.82±1.08	82.94±2.46	78.96±5.13	80.35±3.39	79.22±1.59	78.93±2.68	79.99±0.90	77.17±1.56	77.14±1.60	79.24±2.00	**80.69±1.83**
κ*100	72.28	91.21±0.27	84.49±2.59	85.91±3.79	86.33±2.42	91.76±0.86	91.03±1.37	92.42±0.49	91.39±0.84	93.03±0.48	93.28±0.52	**93.89±0.52**

TABLE XV

DETAILED CLASSIFICATION ACCURACY OF WHU-HI-HANCHUAN ON DIFFERENT NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	SVM	RF	PyResNet	ContextualNet	SSRN	SSFTT	MATNet	SS-MSDCNet	SPTNet	CSCANet	MCTGCL	SSPNet
1	99.56	98.77±0.06	89.44±11.08	93.51±1.66	93.70±2.49	93.66±1.65	96.85±0.54	98.77±0.64	99.08±0.21	99.30±0.10	99.54±0.21	**99.71±0.06**
2	89.55	91.99±0.38	93.16±2.62	91.05±3.33	91.16±5.01	93.90±2.54	94.74±2.90	95.71±1.80	94.95±1.32	96.31±0.95	96.20±0.56	**96.90±0.63**
3	74.49	88.60±1.12	83.72±10.25	85.12±4.44	88.37±6.33	93.33±2.53	95.86±2.42	98.46±1.35	97.36±0.60	97.43±0.64	**99.76±0.48**	99.10±0.35
4	89.87	97.55±0.29	93.67±5.57	93.57±5.48	94.36±6.98	98.25±1.65	97.92±1.10	**99.53±0.51**	98.23±0.56	99.35±0.25	98.89±0.29	99.30±0.36
5	0.00	4.51±1.48	83.28±9.73	62.56±18.94	41.08±36.02	81.58±9.26	76.82±16.78	**95.98±1.66**	90.39±3.52	95.10±1.20	92.36±1.65	94.76±0.96
6	0.80	9.84±1.86	67.41±12.86	61.92±9.39	64.29±11.57	78.47±6.72	80.11±9.08	82.75±7.00	82.28±3.57	80.38±1.59	85.70±1.78	**89.68±1.19**
7	90.43	89.83±0.81	74.65±10.95	81.82±9.75	82.02±9.24	84.52±7.85	88.62±9.05	93.33±2.65	92.90±1.71	**94.43±1.20**	93.18±0.51	93.60±0.92
8	90.77	86.88±0.60	88.21±7.96	84.16±3.96	85.99±4.31	90.48±2.99	93.27±3.75	94.63±2.75	95.38±0.77	95.46±0.99	**97.32±0.49**	97.09±0.66
9	35.56	72.87±1.50	78.87±6.74	77.68±8.08	82.80±5.57	84.96±6.83	88.75±7.40	96.02±1.72	95.25±1.84	96.80±0.55	**98.06±0.44**	97.33±0.87
10	77.06	89.90±1.19	92.76±12.83	96.69±1.90	98.23±1.21	98.04±0.98	98.62±0.87	98.40±0.60	98.55±0.59	98.55±0.29	**99.44±0.08**	98.87±0.13
11	89.60	98.05±0.18	81.53±11.11	92.23±4.24	93.25±2.42	94.19±2.14	93.99±3.04	95.62±4.36	98.37±0.41	**99.09±0.23**	98.77±0.25	99.07±0.24
12	0.03	29.66±1.76	75.41±6.90	59.89±8.07	43.83±23.68	81.87±5.11	85.14±5.75	85.95±9.71	87.84±7.11	94.51±1.40	95.04±2.02	**97.48±1.71**
13	46.12	56.51±0.93	77.38±5.30	72.15±12.51	74.63±11.45	84.06±4.45	86.61±4.83	78.11±13.64	85.15±2.33	84.99±1.34	87.14±1.34	**88.91±1.04**
14	85.06	91.19±0.28	84.30±6.62	87.17±1.93	87.44±5.96	93.55±2.73	95.55±3.50	97.29±1.20	95.76±0.71	96.49±0.36	97.11±0.26	**97.40±0.41**
15	0.00	0.78±0.44	78.76±17.11	79.37±11.56	67.95±39.47	**88.63±14.55**	78.84±17.47	64.36±6.12	82.72±4.46	84.53±3.37	84.33±2.18	84.52±1.51
16	99.78	98.93±0.11	98.51±1.13	99.34±0.50	99.86±0.10	99.77±0.07	99.51±0.57	99.88±0.07	99.74±0.11	99.70±0.06	**99.91±0.04**	99.90±0.03
OA (%)	85.79	89.92±0.09	88.70± 3.74	90.69±1.02	91.58±1.38	94.02±0.57	95.36±1.34	96.63±1.24	96.91±0.33	97.37±0.15	97.87±0.05	**98.15±0.17**
AA (%)	60.54	69.12±0.19	83.82± 2.88	82.39±2.10	80.56±5.70	89.95±1.20	90.70±3.34	92.18±2.14	93.37±0.96	94.53±0.31	95.11±0.19	**95.85±0.34**
κ*100	83.20	88.15±0.11	86.72±4.47	89.09±1.19	90.13±1.63	93.00±0.66	94.57±1.57	96.05±1.46	96.38±0.39	96.92±0.18	97.51±0.05	**97.83±0.19**

experimental results from these scenarios with those obtained using our proposed method.

The results in Table VIII demonstrate that leveraging TCA_Low to extract low-frequency branch spatial features and TCA_High to extract high-frequency branch spatial features is beneficial (as evidenced by the Only_Low, Only_High, and Ours scenarios). Additionally, by comparing Only_Low and TCA_Low, we observed that using the low-frequency branch's TCA_Low operation to extract high-frequency spatial features leads to the loss of spatial details in high-frequency features, thereby impairing classification performance. Meanwhile, comparing Only_High and TCA_High reveals that using the high-frequency branch's TCA_High operation to extract low-frequency spatial features results in nearly unchanged classification accuracy.

3) Modules of SLP: To validate the structural design proposed in our SLP, we conducted ablation analyses on SPB, ALA, and APW across three distinct datasets. Specifically,

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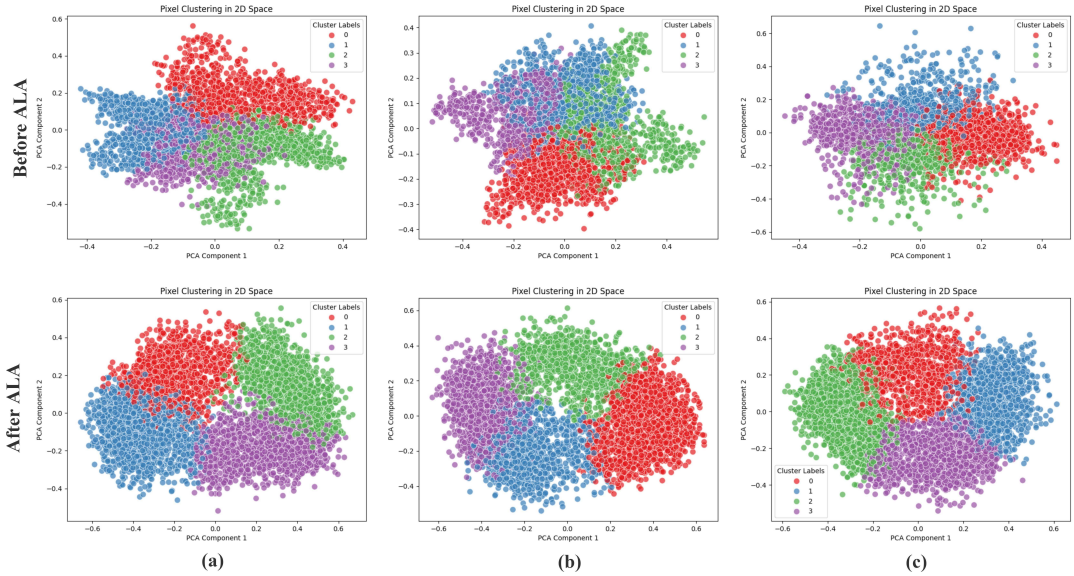

Fig. 8. Visualization of feature distribution after PCA and k-means. (a) Holden Crater. (b) Nili Fossae. (c) Utopia Planitia.

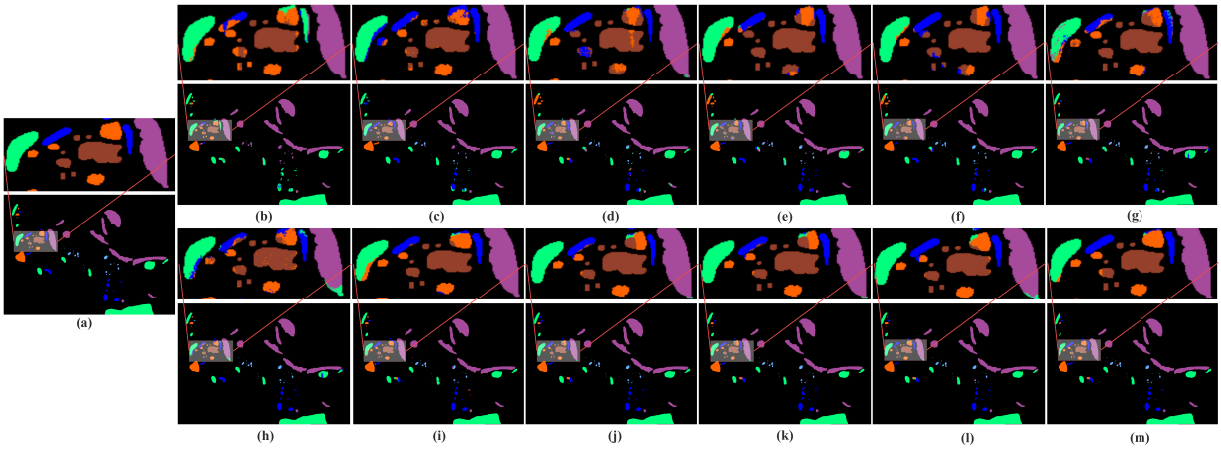

Fig. 9. Visualization of different networks on the Holden Crater dataset. (a) Ground truth. (b) SVM [7]. (c) RF [10]. (d) PyResNet [50]. (e) ContextualNet [51]. (f) SSRN [52]. (g) SSFTT [53]. (h) MATNet [54]. (i) SS-MSDCNet [55]. (j) SPTNet [56]. (k) CSCANet [57]. (l) MCTGCL [48]. (m) Proposed SSPNet.

we individually removed SPB (No_SPB), ALA (No_ALA), and APW (No_APW) from the model on each dataset and compared the experimental results with those of the original model (Ours). The results presented in Table IX demonstrate that across all three datasets, our designed modules have led to varying degrees of improvement in classification performance, with the SLP structure (Ours) achieving the optimal classification performance.

4) SFP and Spectral Linear Perceptron: To investigate the contributions of different perceptual modules to classification performance, we conducted ablation experiments on the Nili Fossae dataset. Specifically, we designed three scenarios: 1) simultaneously removing both SFP and SLP (No_SFP_SLP); 2) removing only SFP (No_SFP); and 3) removing only SLP (No_SLP). The experimental results are presented in Table X. Notably, the scenario with both SFP and SLP removed yielded the poorest performance. Classification accuracy improved significantly when either SFP or SLP was reintroduced, with the SSPNet (the method incorporating both SFP and SLP) achieving the best performance.

Furthermore, to analyze how different perceptual modules affect operational efficiency, we compared the number of parameters, average inference time, and FLOPs across all scenarios. Our findings show that adding SFP and SLP led to minimal changes in parameter count and FLOPs, but inference time increased as the model complexity grew. Overall, we conclude that the runtime overhead is justified by the significant performance improvements brought by the designed perceptual modules.

E. Efficiency Analysis

To enable a more comprehensive comparison with other deep learning-based models, we analyzed model efficiency on the Nili Fossae dataset using three metrics: model parameter count, average inference time, and number of FLOPs. The results are presented in Table XI. Although our proposed method does not achieve optimal performance across all three metrics, it attains a relatively optimal balance in operational efficiency compared to other state-of-the-art

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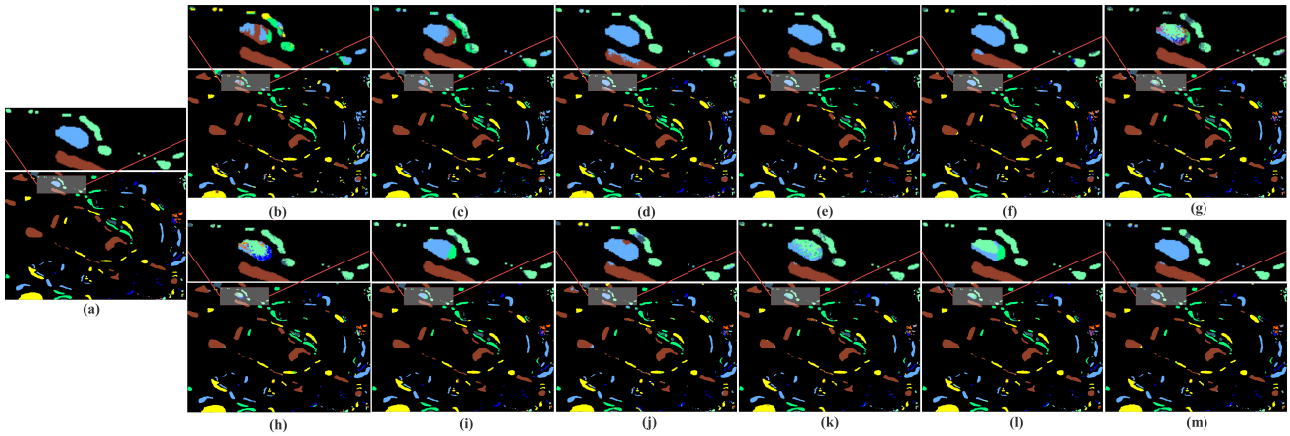

Fig. 10. Visualization of different networks on the Nili Fossae dataset. (a) Ground truth. (b) SVM [7]. (c) RF [10]. (d) PyResNet [50]. (e) ContextualNet [51]. (f) SSRN [52]. (g) SSFTT [53]. (h) MATNet [54]. (i) SS-MSDCNet [55]. (j) SPTNet [56]. (k) CSCANet [57]. (l) MCTGCL [48]. (m) Proposed SSPNet.

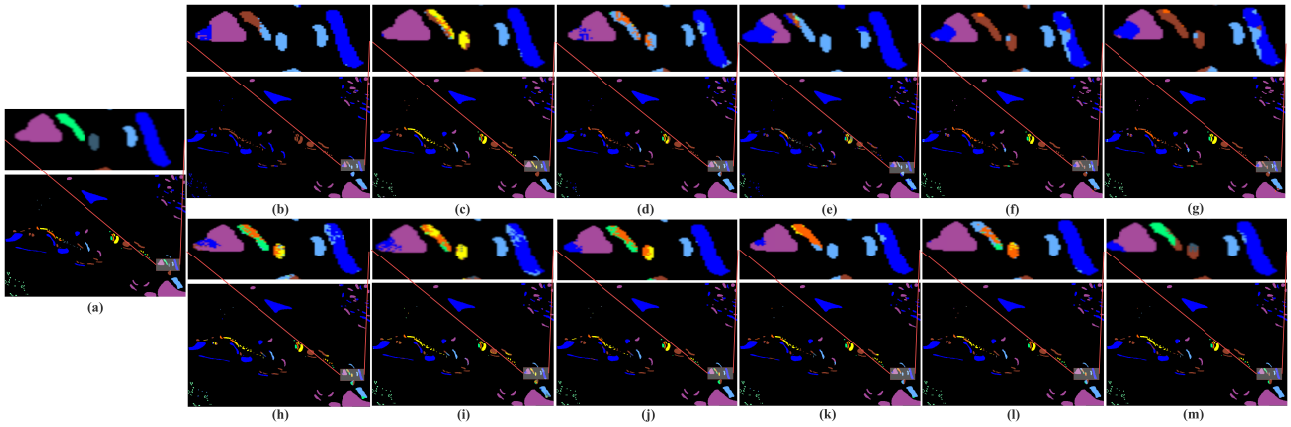

Fig. 11. Visualization of different networks on the Utopia Planitia dataset. (a) Ground truth. (b) SVM [7]. (c) RF [10]. (d) PyResNet [50]. (e) ContextualNet [51]. (f) SSRN [52]. (g) SSFTT [53]. (h) MATNet [54]. (i) SS-MSDCNet [55]. (j) SPTNet [56]. (k) CSCANet [57]. (l) MCTGCL [48]. (m) Proposed SSPNet.

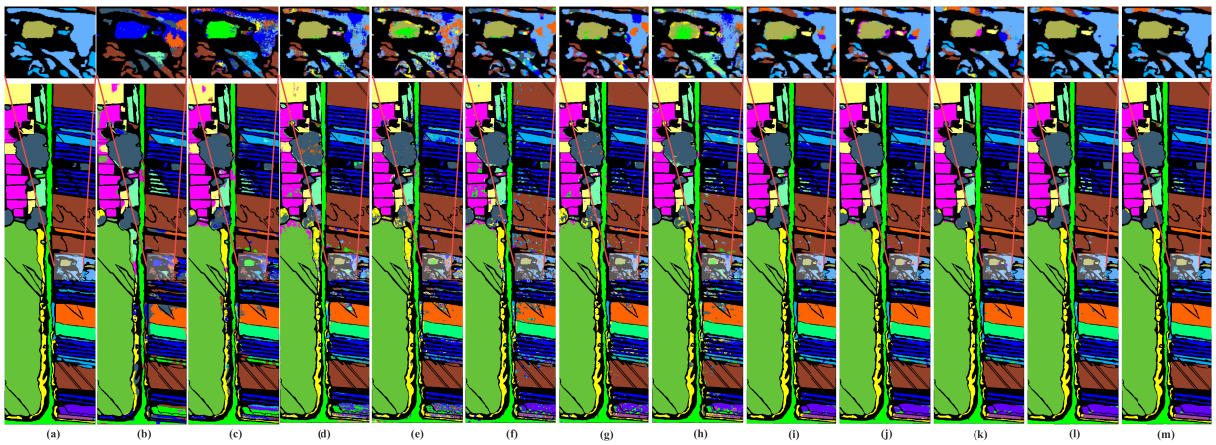

Fig. 12. Visualization of different networks on WHU-Hi-HanChuan dataset. (a) Ground truth. (b) SVM [7]. (c) RF [10]. (d) PyResNet [50]. (e) ContextualNet [51]. (f) SSRN [52]. (g) SSFTT [53]. (h) MATNet [54]. (i) SS-MSDCNet [55]. (j) SPTNet [56]. (k) CSCANet [57]. (l) MCTGCL [48]. (m) Proposed SSPNet.

approaches—particularly under the condition of achieving the best classification performance.

F. Experimental Results and Visualization

We conducted comprehensive experiments on three mineral datasets and one real-world agricultural dataset to perform a comparative evaluation between the proposed method and state-of-the-art approaches. As demonstrated in the experimental results presented in Tables XII–XV, our method achieved superior classification performance across all three evaluation metrics for every dataset.

We further observed that machine learning-based methods tend to underperform on these classes when the number of training samples is limited, thereby resulting in lower

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average precision. Consequently, such methods are ill-suited for scenarios with an imbalanced sample distribution and insufficient training samples. Conversely, deep learning-based methods tend to exhibit greater potential under such circumstances. Examination of the average precision across all datasets reveals that all deep learning-based methods outperform machine learning-based methods on this metric. Additionally, our proposed SSPNet consistently outperforms prevailing techniques in terms of classification accuracy and robustness under diverse environmental conditions.

To more intuitively demonstrate how our designed ALA mechanism enhances feature distinguishability between similar categories, we compress the features before and after using ALA into a 2-D space using the same PCA dimensionality reduction method and apply the same k-means clustering technique to observe the distribution of different features, as shown in Fig. 8. On three mineral datasets, we can clearly observe that after applying ALA, the previously hard-to-distinguish feature distributions are significantly improved. Our method makes these similar features more distinguishable, thereby facilitating the final classification results. In addition, to more clearly analyze the classification improvement, the visualized results of classification are shown in Figs. 9–12. Our proposed method also achieves the best classification performance in the visualization of all datasets.

V. Conclusion

To address the classification challenges caused by spatial–spectral feature similarities between similar materials, we designed a spatial–spectral perception network (SSPNet). The SSPNet consists of two main stages: the spatial–frequency perception stage and the spectral linear perception stage. The spatial–frequency perceiver we designed projects hard-to-differentiate spatial features into low- and high-frequency spaces using the Fourier transform. Additionally, our proposed TCA is used to extract low- and high-frequency 3-D spatial features, enabling more detailed extraction of spatial–frequency features in HSIs. Furthermore, in the spectral linear perception stage, we designed a flexible SPB, which includes an absolute position-weighted multihead linear self-attention mechanism (ALA). ALA not only facilitates attention interactions between features at different scales but also incorporates the significant impact of absolute positions between different spectra in HSIs. This strengthens the classification influence of neighboring spectra, further enhancing classification performance under complex conditions. Our proposed method achieves superior classification performance over the state-of-the-art methods with both three real mineral scenarios and one agricultural scenario dataset.

Although the proposed method achieves ideal classification performance for most categories, it performs poorly for certain classes with very few samples (such as class No.5 in Nili Fossae and class No.8 in Utopia Planitia). This leads to a decrease in classification performance, which, in turn, affects the AA metric. In the future, we plan to design more methods for HSI classification with limited samples, building upon our current work to better address practical real-world requirements.

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